Problem

Title

LCM of N Numbers (Without Using Library Functions)

Topics

LCM, GCD, Euclidean Algorithm, Loops, Nested Loops, Mathematical Logic

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

During a placement preparation session, students are given a mathematical challenge involving multiple numbers.

The trainer provides an integer **N**, followed by **N positive integers**. The task is to compute the **Least Common Multiple (LCM)** of all the given numbers.

The **Least Common Multiple (LCM)** of two or more numbers is the smallest positive integer that is divisible by all of them.

For example:

- LCM of 4 and 6 = 12
- LCM of 2, 3, 4 = 12
- LCM of 5, 10, 15 = 30

Students are NOT allowed to:

- Use built-in LCM or GCD library functions.
- Use BigInteger libraries.

They must:

1. Understand the mathematical relationship:
2. `LCM(a, b) = (a × b) / GCD(a, b)`
3. Implement GCD using the Euclidean Algorithm (loop-based).
4. Iteratively compute LCM of all numbers.
5. Handle large intermediate values carefully.
6. Use proper looping and condition structure.

Edge cases to consider:

- If any number is 0 → LCM should be 0.
- If N = 1 → LCM is the number itself.
- Large values may cause overflow (students should think carefully).

This problem strengthens:

- Mathematical reasoning
- Iterative accumulation logic
- Nested function implementation
- Loop-based Euclidean algorithm
- Conditional edge-case handling

Input Format

```
N
a1 a2 a3 ... aN
```

Constraints:

```
1 ≤ N ≤ 1000
0 ≤ ai ≤ 10^6
```

Output Format

- Print a single integer representing the LCM of all given numbers.

Examples

Example 1

Input:

```
3
2 3 4
```

Output:

```
12
```

Example 2

Input:

```
2
4 6
```

Output:

```
12
```

Example 3

Input:

```
4
5 10 15 20
```

Output:

```
60
```

Example 4

Input:

```
1
7
```

Output:

```
7
```